State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3

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Abstract

[INTERNAL THEOREM] We prove, unconditionally, that the Connes–van Suijlekom truncations of the round $S^3 = \mathrm{SU}(2)$ Camporesi–Higuchi spectral triple converge to the continuum spectral triple in van Suijlekom's state-space Gromov–Hausdorff distance [8], with explicit rate $d_{\mathrm{GH}} \leq \gamma_{n_{\max}} = (4/\pi + o(1)) \log n_{\max}/n_{\max}$ in the dual-Coxeter normalisation of the SU(2) metric (equivalently $2/\pi$ on the unit round S^3; the constant scales with the metric, 2026-09-03 convention note). **[INTERNAL THEOREM]** The truncated state spaces are metrized by the left-translation Lipschitz seminorm, which equals the Dirac/metric Lipschitz constant on the continuum and whose kernel at every finite cutoff is exactly the scalars (the truthful Dirac-commutator seminorm itself degenerates at finite cutoff; **[PANEL-VERIFIED]** a remark quantifies this on a finite panel of cutoffs). The proof is a compression/lifted-state argument in the pattern of the compact-metric-group theorem of Gaudillot-Estrada–van Suijlekom [7], with self-contained proofs: the dual map evaluates conjugated operators against the central spectral Fejér vector state, which embeds *exactly* into the spinor window via the $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ block decomposition $V_j \otimes (V_{j\pm 1/2})$ of the Camporesi–Higuchi eigenspaces, and both almost-inverse defects collapse to Fejér smoothing — one on functions, one as operator conjugation-averaging — at the common moment $\gamma_{n_{\max}}$. The contribution relative to the scalar theory: the spinor-window transport, per-band multiplicity-one injectivity on the chirality-doubled multiplier system, the closed-form rate moment, and the $4/\pi$ asymptotic constant. Supporting structure is organized in five lemmas (the fifth since withdrawn 2026-09-03): (L1′) a truncated operator system on the full chirality-doubled spinor bundle; (L2) **[INTERNAL THEOREM + MEASURED]** a positive central spectral Fejér kernel on SU(2) with closed-form mass-concentration moment and Stein–Weiss-type quantitative rate $\gamma_n \sim (4/\pi) \log n/n$ (dual-Coxeter normalisation; $2/\pi$ on the unit S^3); (L3) **[PANEL-VERIFIED]** a Lipschitz comparison bound $\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \leq C_3 \|\nabla f\|_{L^\infty}$ with $C_3 = 1$ (verified on a finite harmonic panel rather than proved in general, and *not* used by the unconditional theorem); (L4) a Berezin map (convolution form) that is positive, contractive, an approximate identity, and L3-compatible; (L5, since *withdrawn* 2026-09-03: the bound is established by the unconditional lifted-state theorem, not this approximation-pair assembly). **[OBSERVATION]** The asymptotic constant is a quotient of sphere volumes — $4/\pi$ in the dual-Coxeter normalisation of §2.1, $2/\pi = \mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ at unit radius, and *not* the Hopf fibration's base-to-total ratio (§3.2) — exhibiting the expected one-log slowdown of the non-abelian SU(2) rate relative to the Leimbach–van Suijlekom [17] torus rate $O(1/n_{\max})$. State-space Gromov–Hausdorff convergence of scalar truncations for *all* compact metric groups — including SU(2) — is established by Gaudillot-Estrada–van Suijlekom [7]; the residual contribution here is the chirality-doubled Camporesi–Higuchi spinor substrate, the closed-form rate moment, and the explicit $4/\pi$ asymptotic constant. The construction supplies quantitative metric data for almost-commutative spectral-triple constructions hosted on S^3 [18].

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1 Introduction

A central problem in noncommutative metric geometry is whether finite-dimensional truncations of a Riemannian spectral triple converge, in a suitable quantum Gromov–Hausdorff sense, to the continuum geometry. Connes and van Suijlekom [4] introduced the framework of *truncated operator systems* $\mathcal{O}_\Lambda = P_\Lambda \mathcal{A} P_\Lambda$ obtained by spectral projection onto the modes of D with $|D| \leq \Lambda$, and showed that the natural propagation invariant $\mathrm{prop}(\mathcal{O}_\Lambda) = 2$ is realised on the Toeplitz operator system on S^1. They left explicit convergence to the continuum in a quantitative quantum Gromov–Hausdorff distance as an open question, deferring it to "elsewhere." The convergence itself, at the state-space level, is established in van Suijlekom [8] for the circle and sphere, and for *all compact metric groups* by Gaudillot-Estrada–van Suijlekom [7]; the residual contribution of the present paper is therefore the spinor substrate, the closed-form rate moment, and the explicit $4/\pi$ constant — not the existence of convergence for SU(2) scalars.

The flat-structure case has been worked out in several subsequent papers. Hekkelman [6] treated the circle S^1 with the Toeplitz/Fejér kernel reconstruction. Leimbach and van Suijlekom [17] proved state-space GH convergence of truncated tori with explicit rate $O(1/\Lambda)$, using the standard Schur-multiplier characterisation of the abelian-compact-group Fejér mechanism. A complementary line of work treats *Berezin–Toeplitz* reconstruction on S^2, using the Kähler structure of the coadjoint orbit (see [9] and references therein). All of these results address flat abelian or Kähler structures.

Adjacent activity in the broader propinquity program addresses orthogonal corners of the same general framework. Toyota [13] establishes quantum-Gromov–Hausdorff convergence of spectral truncations for discrete groups of polynomial growth, a scope which excludes compact Lie groups by definition. Farsi–Latrémolière [10] prove spectral continuity for collapse of spectral triples along vertical directions in the spectral propinquity, a collapsing-family setup distinct from the fixed-manifold truncation sequence here. Farsi–Latrémolière [11] establish continuity of the spectral propinquity along analytic paths of Riemannian metrics on a fixed closed spin manifold, a continuous-family setup distinct from the discrete-truncation sequence considered here. Hekkelman–McDonald [12] develop a noncommutative integral and a Szegö limit formula for spectrally truncated triples in the Connes–van Suijlekom paradigm, a companion direction complementary to the metric-side convergence proved here.

For the most natural physically motivated non-Kähler compact case — the round $S^3 = \mathrm{SU}(2)$ — the scalar state-space convergence is covered by the compact-metric-group theorem of Gaudillot-Estrada–van Suijlekom [7]; what remains distinctive is the *Dirac spectral triple* version on the Camporesi–Higuchi spinor substrate, with explicit closed-form rate.

[INTERNAL THEOREM] The present paper treats that case; the theorem is unconditional under the translation-seminorm metrization of § 1.1. Let $\mathcal{T}_{S^3} = (C^\infty(S^3), L^2(S^3, \Sigma), D_{\mathrm{CH}})$ denote the *Camporesi–Higuchi spectral triple*, with D_{CH} the Dirac operator on the unit round S^3 in the spinor harmonic basis of Camporesi–Higuchi [2], with spectrum $\sigma(D_{\mathrm{CH}}) = \{\pm(n + 1/2) : n \in \mathbb{N}\}$ on each chirality sector.[1] Let $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}})$ denote its truncation at cutoff $n_{\max}$, where $\mathcal{O}_{n_{\max}}$ is the truncated operator system in the sense of Connes–van Suijlekom and $D_{n_{\max}}$ is the restriction of D_{CH} to $\mathcal{H}_{n_{\max}} = P_{n_{\max}} L^2(S^3, \Sigma)$.

[1] We use the Fock-shell labelling $n_{\mathrm{Fock}} = n_{\mathrm{CH}} + 1$ throughout, so the lowest non-trivial level is $n = 1$ with $|\lambda_1| = 3/2$. This convention is fixed by [2, §2] and matches the SO(4) representation-theoretic indexing used in the supporting computational modules of the present work.

Theorem 1.1 (Main theorem; cf. §1.1 and §4). ***[INTERNAL THEOREM]*** *Unconditionally, the truncated triples $\mathcal{T}_{n_{\max}}$ converge to $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance [8], under the translation-seminorm metrization of § 1.1, with explicit rate*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; \gamma_{n_{\max}}, \qquad \gamma_{n_{\max}} = \frac{4\log n_{\max}}{\pi\, n_{\max}} + O\big(1/n_{\max}\big) \quad \text{(dual-Coxeter normalisation; } 2/\pi \text{ on the unit } S^3 \tag{1}$$

where $\gamma_{n_{\max}}$ is the mass-concentration moment of the central spectral Fejér kernel on SU(2) *given by Lemma 3.4. The unconditional statement is Theorem 1.7, whose bound is $\le \gamma_{n_{\max}}$ outright and does not invoke C_3; the main theorem's display now states that bound (until 2026-09-04 it carried a $C_3 = 1$ factor inherited from the withdrawn Lemma 3.12 route).* ***[OBSERVATION]*** *The asymptotic constant is $4/\pi$ in the dual-Coxeter normalisation of §2.1 and $2/\pi$ on the unit S^3; it is the sphere-volume quotient $2\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ for unit-radius spheres — though not the Hopf base-to-total ratio (see the volume-ratio note in §3.2) (*metric convention, 2026-09-03: *$\gamma_{n_{\max}}$ as computed in Lemma 3.4 weights the kernel by the* SO(3) *rotation angle $\chi \in [0, 2\pi]$, which is* twice *the geodesic distance $\theta = \chi/2$ on the unit round S^3; the constant is $4/\pi$ in that dual-Coxeter normalisation — the round S^3 of radius 2 — and $2/\pi$ on the unit S^3 (the $d_1 = d/2$ metric of §2.1), the two differing by the metric scale, which enters a distance moment linearly; the theorem holds in either normalisation,* `tests/test_p38_metric_convention.py`*).*

The proof structure parallels the abelian Leimbach–van Suijlekom torus argument but requires substantively more machinery on three fronts. First, the central spectral Fejér kernel on SU(2) must be constructed using Peter–Weyl decomposition rather than Schur–Fourier multipliers; the natural-coefficient version (carrying the $\sqrt{2j+1}$ Plancherel weight) gives a Cesàro-2-style $\log n/n$ rate rather than the unweighted $1/n$ rate of the torus. Second, the Lipschitz comparison estimate on the round S^3 requires the Wen–Avery 3-Y integral [1, 25] to control the multiplier matrices in the truncated basis, and depends on the SO(4) selection rule $|n - n'| + 1 \le N \le n + n' - 1$ for non-vanishing 3-Y matrix elements. Third, the Berezin reconstruction map on SU(2) does not arise from a Kähler structure (the round S^3 has no compatible almost-complex structure), so we use the central-Fejér-kernel spectral form directly, which is the natural non-Kähler analog of the Connes–van Suijlekom S^1 Fejér reconstruction.

The asymptotic constant in (1) carries geometric content, and its value is normalisation-dependent (2026-09-03): in the dual-Coxeter normalisation of §2.1 the constant is $4/\pi$, which coincides numerically with $\mathrm{Vol}(S^2)/\pi^2$ for $\mathrm{Vol}(S^2) = 4\pi$; on the unit S^3 it is $2/\pi$, the factor 2 being the metric scale rather than a geometric signature. *Which volume ratio (2026-09-03).* The constant is a genuine quotient of sphere volumes — but not the one its usual name suggests. For unit-radius spheres $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 4\pi/2\pi^2 = 2/\pi$ exactly, and the dual-Coxeter constant is twice that. It is *not* the Hopf fibration's base-to-total ratio: the Hopf map is a Riemannian submersion $S^3(r) \to S^2(r/2)$, so that ratio is $1/(2\pi)$ on the unit sphere and $1/(4\pi)$ in the dual-Coxeter metric. The *volume content* of the constant — what places it in the $k = 0$ (M1) slot of the master Mellin engine and in $\mathbb{Q}[\pi, \pi^{-1}]$ — is therefore intact; only the "Hopf base" label is a misnomer, that base carrying half the radius of the sphere whose volume appears. Separately, **[OBSERVATION]** the same ratio $2/\pi$ is the first-moment constant of the classical Fejér kernel on the circle (Remark 3.5). This is the volume factor that the $k = 0$ (trivial) heat kernel on the round S^3 leaves behind — historically called the "Hopf-base measure", though it is a quotient of unit-sphere volumes rather than the Hopf fibration's base-to-total ratio (§3.2) — and which has been independently identified in the internal supporting analysis as one of the three sub-mechanisms of a master Mellin-engine $\mathrm{Tr}(D^k e^{-tD^2})$ with operator order $k \in \{0, 1, 2\}$ [26]. **[OBSERVATION]** The constant in (1) ($2/\pi$ on the unit S^3; $4/\pi$ in the dual-Coxeter normalisation) sits in the $k = 0$ sub-case (M1, the volume slot) of that classification; the same $1/\pi^2$ factor reappears in the Pythagorean Hilbert–Schmidt orthogonality corollary on the Krein wedge documented in the companion Lorentzian-extension paper (Paper 43 [30]). **[OBSERVATION]** The appearance of this constant in the state-space GH rate is therefore

not an isolated occurrence: the rate at which the framework recovers its continuum limit carries the same transcendental signature that the framework's heat-kernel machinery imprints on its bulk observables. No derivation links the two appearances; the shared value is the whole of the evidence.

Outline. Section 2 sets up the round S^3 Camporesi–Higuchi spectral triple, its truncations, and the Peter–Weyl basis. Section 3 proves lemmas L1′–L4; Lemma 3.12 is *withdrawn* [retracted 2026-09-05: l5-assembly-listed-as-live-contribution]. Each surviving lemma is self-contained: Lemma 3.1 treats the chirality-doubled operator system substrate; Lemma 3.4 the central spectral Fejér kernel and its quantitative rate; Lemma 3.7 the Lipschitz comparison bound with $C_3 = 1$; Lemma 3.10 the Berezin reconstruction. Section 4 assembles the main theorem via the lifted-state route. Section 5 discusses the role of the $4/\pi$ constant, the Marcolli–van Suijlekom [18] gauge-network framework that the result anchors, the obstruction to extending the proof to the holomorphic-sector S^5 Bargmann–Segal case, and open questions.

1.1 The unconditional theorem: compression and lifted states under the translation seminorm

Both gaps that conditioned earlier formulations of the main theorem are closed by one reframing: metrize the truncated state space by the *left-translation Lipschitz seminorm* rather than by the truthful-Dirac commutator seminorm. On the continuum the two agree exactly (Lemma 1.3); at finite cutoff the translation seminorm is non-degenerate while the Dirac-commutator seminorm is not (Remark 1.9). All proofs in this subsection are self-contained; the kernel quantities $K_{n_{\max}}$, $Z_{n_{\max}}$, $\gamma_{n_{\max}}$ are those of § 3.2 below (forward references).

Definition 1.2 (translation seminorm)**.** Let λ_g denote left translation on $S^3 = \mathrm{SU}(2)$, acting on the spinor space $L^2(S^3, \Sigma) \cong L^2(\mathrm{SU}(2)) \otimes \mathbb{C}^2$ (left trivialization) by $U_g = \lambda_g \otimes 1$, and $\rho_g(T) = U_g T U_g^*$. Since D_{CH} is built from the left-invariant frame, $[U_g, D_{\mathrm{CH}}] = 0$; hence U_g preserves every truncation $\mathcal{H}_{n_{\max}}$ and conjugation preserves the operator system: $\rho_g(P_{n_{\max}} M_f P_{n_{\max}}) = P_{n_{\max}} M_{\lambda_g f} P_{n_{\max}}$. For $T \in \mathcal{O}_{n_{\max}}$ and $f \in C(S^3)$ define

$$L_{n_{\max}}(T) := \sup_{g \neq e} \frac{\|\rho_g(T) - T\|_{\mathrm{op}}}{d(e,g)}, \qquad L(f) := \sup_{g \neq e} \frac{\|\lambda_g f - f\|_{L^\infty}}{d(e,g)}, \tag{2}$$

with d the dual-Coxeter geodesic distance of §2.1 — equivalently the rotation angle, $d(e,g) = \chi(g)$ — which is bi-invariant.

Lemma 1.3 (continuum identification)**.** *For $f \in C^\infty(S^3)$, $L(f) = \mathrm{Lip}_d(f) = \|\nabla_d f\|_{L^\infty} = \frac{1}{2}\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}}$, all four quantities taken in the dual-Coxeter metric d of §2.1 except the last, in which D_{CH} is the unit-metric operator (Eq. (6)).*

Proof. $\|\lambda_g f - f\|_\infty = \sup_x |f(g^{-1}x) - f(x)| \leq \mathrm{Lip}(f)\, d(g^{-1}x, x) = \mathrm{Lip}(f)\, d(e,g)$ by bi-invariance, so $L(f) \leq \mathrm{Lip}_d(f)$; conversely left translations act transitively and realize every short geodesic ($d(g^{-1}x, x) = d(e,g)$ along the orbit), so the supremum recovers $\mathrm{Lip}_d(f)$. The identity $\mathrm{Lip}_d(f) = \|\nabla_d f\|_\infty$ on the continuum is standard, as is $\mathrm{Lip}_{d_1}(f) = \|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}}$ for the unit metric (Connes' distance formula on a closed spin manifold); since $d = 2d_1$, Lipschitz seminorms scale inversely and the factor $\frac{1}{2}$ of (6) follows. □

Lemma 1.4 (per-band injectivity)**.** *For $1 \leq N \leq 2n_{\max} - 1$ let $E_N \subset C^\infty(S^3)$ be the degree-$(N-1)$ harmonic band ($\dim E_N = N^2$), an irreducible $\mathrm{SO}(4)$-module of type $V_{(N-1)/2} \otimes V_{(N-1)/2}$. The compression $\iota_N : E_N \to B(\mathcal{H}_{n_{\max}})$, $f \mapsto P_{n_{\max}} M_f P_{n_{\max}}$, is injective.*

Proof. ι_N is $\mathrm{SO}(4)$-equivariant for the conjugation action (rotations implemented on $\mathcal{H}_{n_{\max}}$ commute with $P_{n_{\max}}$). Since E_N is irreducible, ι_N is either zero or injective (Schur). It is

nonzero: every multiplier matrix M_{NLM} with $N \le 2n_{\max}-1$ is nonzero, by non-vanishing of the stretched 3-Y matrix element within the top kept shell (Wen–Avery closed form [1]); **[PANEL-VERIFIED]** verified symbolically/numerically for every band at $n_{\max} \le 5$, where the per-band rank additionally equals the full N^2 (`debug/p38_g1g2_band_diagnostics.py`, frozen in `tests/test_p38_action_seminorm.py`). □

Proposition 1.5 (kernel condition on the truthful substrate). *$L_{n_{\max}}(T) = 0$ if and only if $T \in \mathbb{C}1$.*

Proof. $L_{n_{\max}}(T) = 0$ means $\rho_g(T) = T$ for all g. Write $T = P_{n_{\max}} M_f P_{n_{\max}}$ with f band-limited to $N \le 2n_{\max}-1$ (every element of $\mathcal{O}_{n_{\max}}$ has this form). Then $P_{n_{\max}} M_{\lambda_g f - f} P_{n_{\max}} = 0$; left translation preserves each harmonic band, so $\lambda_g f - f$ is band-limited and Lemma 1.4 gives $\lambda_g f = f$ for all g; transitivity forces f constant. The converse is immediate. Finite dimensionality plus the kernel condition make $(S(\mathcal{O}_{n_{\max}}), \mathrm{MK}_{L_{n_{\max}}})$ a compact metric space of finite diameter, and Lemma 1.3 provides the same on the continuum side — the standard Monge–Kantorovich setting on both ends. □

Lemma 1.6 (spinor window inclusion and the lifted state). *Let $J = (n_{\max}-1)/2$, let $h = \sum_{j \le J} \sqrt{2j+1}\, \chi_j$ be the central spectral Fejér amplitude of Definition 3.3 (the scalar window then has exactly $n_{\max}$ levels, and $|h|^2/Z_{n_{\max}}$ is the Fejér kernel $K_{n_{\max}}$), fix a unit spinor $\chi \in \mathbb{C}^2$, and set $\xi := (h \otimes \chi)/\sqrt{Z_{n_{\max}}}$. Then:*

*(a) (*Exact window inclusion.*) $\xi \in \mathcal{H}_{n_{\max}}$. Under $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$,*

$$L^2(\mathrm{SU}(2)) \otimes \mathbb{C}^2 \;=\; \bigoplus_j V_j \otimes \big(V_{j+1/2} \oplus V_{j-1/2}\big),$$

and these summands are precisely the Camporesi–Higuchi eigenspaces [2]: in the level index of (4), $V_j \otimes V_{j+1/2}$ is the positive shell $n = 2j+1$ and $V_j \otimes V_{j-1/2}$ the negative shell $n = 2j$, with dimensions $n(n+1)$ matching shell by shell. Hence $h \otimes \chi$ lies in the union of the complete *shells $n \le 2J+1 = n_{\max}$ — exact containment, no spillover.*

*(b) (*Lifted state.*) For every $f \in C(S^3)$, $\langle \xi, (P_{n_{\max}} M_f P_{n_{\max}})\, \xi \rangle = \int_{S^3} f\, K_{n_{\max}}\, d\mu_{\mathrm{Haar}}$: the induced measure is exactly the Fejér measure, with first moment $\gamma_{n_{\max}}$ (Lemma 3.4(d)).*

*(c) (*Dual map.*) $\upsilon : \mathcal{O}_{n_{\max}} \to C(S^3)$, $\upsilon(T)(g) := \langle U_g \xi, T\, U_g \xi \rangle$, is unital and positive, satisfies the tautological contraction $L(\upsilon(T)) \le L_{n_{\max}}(T)$, and acts on compressed multipliers as Fejér smoothing: $\upsilon(P_{n_{\max}} M_f P_{n_{\max}}) = K_{n_{\max}} * f$.*

Proof. (a) is the displayed decomposition ($\mathbb{C}^2$ in the right tensor slot; $V_j \otimes V_{1/2} = V_{j+1/2} \oplus V_{j-1/2}$), with the eigenvalue and dimension identifications of Camporesi–Higuchi [2]. (b): $\xi \in \operatorname{ran} P_{n_{\max}}$ absorbs the projections, so the pairing is $\int f\, |h|^2\, |\chi|^2 / Z_{n_{\max}} = \int f\, K_{n_{\max}}$. (c): positivity and unitality are immediate from the vector-state form. The contraction: for $q = g' g^{-1}$, $|\upsilon(T)(g) - \upsilon(T)(g')| = |\langle \xi, (\rho_{g^{-1}}(T) - \rho_{g'^{-1}}(T))\, \xi \rangle| \le \|\rho_q(T) - T\|_{\mathrm{op}} \le L_{n_{\max}}(T)\, d(e,q) = L_{n_{\max}}(T)\, d(g,g')$ by bi-invariance. The smoothing identity: $\upsilon(P M_f P)(g) = \int f(x)\, K_{n_{\max}}(g^{-1}x)\, dx = (K_{n_{\max}} * f)(g)$, using centrality and inversion-invariance of $K_{n_{\max}}$; **[MEASURED]** verified to machine precision ($\le 3 \times 10^{-16}$; the $n_{\max} = 2$ scalar window is frozen in `tests/test_p38_action_seminorm.py`, the across-bands sprint check recorded in the CHANGELOG). □

Theorem 1.7 (Main theorem, unconditional; **[INTERNAL THEOREM]**). *For every $n_{\max} \ge 1$,*

$$d_{\mathrm{GH}}\Big(\big(S(\mathcal{O}_{n_{\max}}), \mathrm{MK}_{L_{n_{\max}}}\big),\ \big(S(C(S^3)), \mathrm{MK}_L\big)\Big) \;\le\; \gamma_{n_{\max}} \;=\; \frac{4}{\pi}\,\frac{\log n_{\max}}{n_{\max}} + O(1/n_{\max}) \qquad (3)$$

in the dual-Coxeter normalisation of Lemma 3.4, i.e. $\gamma_{n_{\max}} = 2 \int K_{n_{\max}}\, d_{\mathrm{round}}\, dg$ on the unit S^3, where the same bound reads $(2/\pi) \log n_{\max}/n_{\max}$ (2026-09-03 convention note; the theorem holds in either normalisation).

Proof. Let $S : C(S^3) \to \mathcal{O}_{n_{\max}}$, $S(f) = P_{n_{\max}} M_f P_{n_{\max}}$ (unital completely positive) and let υ be the dual map of Lemma 1.6. Four facts:

(i) $L_{n_{\max}}(S(f)) \le L(f)$: $\rho_g(S(f)) - S(f) = S(\lambda_g f - f)$ and compression contracts operator norms.

(ii) $L(\upsilon(T)) \le L_{n_{\max}}(T)$ (Lemma 1.6(c)).

(iii) $\|\upsilon(S(f)) - f\|_\infty = \|K_{n_{\max}} * f - f\|_\infty \le \gamma_{n_{\max}} L(f)$ (the good-kernel estimate in the proof of Lemma 3.10(c), at the function level).

(iv) $\|S(\upsilon(T)) - T\|_{\mathrm{op}} \le \gamma_{n_{\max}} L_{n_{\max}}(T)$: on $T = PM_fP$,

$$S(\upsilon(T)) = P\, M_{K_{n_{\max}} * f}\, P = \int_{\mathrm{SU}(2)} \rho_g(T)\, K_{n_{\max}}(g)\, dg =: \Phi(T)$$

— the operator-side conjugation average against the same Fejér measure — and $\|\Phi(T) - T\|_{\mathrm{op}} \le \int \|\rho_g(T) - T\|_{\mathrm{op}}\, K_{n_{\max}}(g)\, dg \le L_{n_{\max}}(T) \int d(e,g)\, K_{n_{\max}}(g)\, dg = \gamma_{n_{\max}} L_{n_{\max}}(T)$.

By (i)–(ii) the state-space pullbacks $S^* : S(\mathcal{O}_{n_{\max}}) \to S(C(S^3))$ and $\upsilon^* : S(C(S^3)) \to S(\mathcal{O}_{n_{\max}})$ are 1-Lipschitz for the Monge–Kantorovich metrics; by (iii)–(iv) they are $\gamma_{n_{\max}}$-almost mutually inverse, e.g. $d(S^*\upsilon^*\omega, \omega) = \sup_{L(f)\le 1} |\omega(\upsilon(S(f))) - \omega(f)| \le \gamma_{n_{\max}}$. The correspondence $\{(\omega, \upsilon^*\omega)\} \cup \{(S^*\phi, \phi)\}$ then has distortion $\le 2\gamma_{n_{\max}}$ (the standard four-pair approximate-isometry computation), whence $d_{\mathrm{GH}} \le \gamma_{n_{\max}}$. The rate is Lemma 3.4(d.ii)–(d.iii). □

Remark 1.8 (no inverse estimate; the Berezin pair as refinement)**.** Both almost-inverse defects in the proof are Fejér smoothings at the same moment $\gamma_{n_{\max}}$ — one on functions, one as the conjugation average Φ — so no partial-inverse or transference estimate is required anywhere: the dual-reach gap of earlier formulations is dissolved rather than closed. The Berezin map $B_{n_{\max}}$ of Definition 3.9 (Lemmas L4–L5) remains as an explicit forward-reconstruction refinement; it is no longer load-bearing for the distance bound. The scalar $C(G)$ case of this compression/lifted-state pattern, for general compact metric groups with the same single-moment bound, is due to Gaudillot-Estrada–van Suijlekom [7]; the contribution here is the spinor-window transport (the exact-fit inclusion of Lemma 1.6(a)), per-band multiplicity-one injectivity on the chirality-doubled system, the closed-form rate moment, and the $4/\pi$ asymptotic constant. All assembly steps above are proved self-contained.

Remark 1.9 (degeneracy of the Dirac-commutator seminorm)**. [MEASURED]** The truthful chirality-diagonal commutator seminorm $\|[D_{\mathrm{CH}}, \cdot]\|_{\mathrm{op}}$ on $\mathcal{O}_{n_{\max}}$ has kernel strictly larger than the scalars at the cutoffs measured (10/14 multipliers at $n_{\max} = 2$, 26/55 at $n_{\max} = 3$, bit-exact; every top-band multiplier compresses shell-diagonally), so it does not metrize the truncated state space. (The frozen falsifier exercises the $n_{\max} = 2$ counts; the $n_{\max} = 3$ counts are measured but not frozen.) An engineered off-diagonal modification of D_{CH} restores its kernel condition (1/14, 1/55) but is no longer needed for the main theorem. Frozen falsifier for the kernel counts: `tests/test_p45_kplus_degeneracy.py::test_spatial_kernel_condition_truthful_vs_offdiag` (`tests/test_p38_action_seminorm.py` backs the injectivity and smoothing premises, not these counts; pointer corrected 2026-09-03).

Remark 1.10 (history)**.** An earlier draft of this paper (May 2026; repository history and the corresponding Zenodo deposit) stated the main theorem in the Latrémolière propinquity (a formulation since withdrawn), with a Lemma L2 that conflated the kernel's Plancherel mass distribution with its multiplier symbol and a dual-reach argument resting on that conflation. A verification pass (2026-06-09; repository `CHANGELOG`, v3.106.0) produced an intermediate conditional form; the present unconditional form (2026-06-10) follows from the translation-seminorm reframing of this subsection. The companion Lorentzian paper proves the related K^+-compression degeneracy theorem.

2 Setup

Metric convention. The distance in which convergence is proved in this paper is van Suijlekom's *state-space Gromov–Hausdorff distance* [8]: whenever we write $\Lambda(\mathcal{T}_1, \mathcal{T}_2)$ we mean $\mathrm{d}_{\mathrm{GH}}^{\mathrm{vS}}(S(\mathcal{O}_1), S(\mathcal{O}_2))$, the Gromov–Hausdorff distance between the state spaces of the associated operator systems in their Monge–Kantorovich metrics; the definition is restated as (25) in §3.5. The strictly stronger *Gromov–Hausdorff propinquity for metric spectral triples* of Latrémolière [15] (Adv. Math. **404** (2022), Paper No. 108393, preprint `arXiv:1811.10843`, 2017) — the spectral-triple-level extension of the dual Gromov–Hausdorff propinquity of [14] (J. Math. Pures Appl. **103** (2015)) and of the original Gromov–Hausdorff propinquity for quantum compact metric spaces of [16] (Trans. AMS **368** (2016)) — is *not* the distance established here: **[OPEN]** its tunnel hypotheses are not verified for the truncated operator system (§3.5), and convergence in that propinquity remains the strictly stronger open target.

2.1 The round S^3 Camporesi–Higuchi spectral triple

Take $G = \mathrm{SU}(2)$ with the bi-invariant metric in the *dual-Coxeter normalisation*: the Ad-invariant inner product on $\mathfrak{su}(2)$ fixed by $\mathrm{Cas}(\mathrm{ad}) = h^\vee = 2$, namely $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$ in the defining representation. On SU(2) this normalisation makes the geodesic distance from the identity equal to the *rotation angle*: for $g = \exp(i\theta\,\hat{n}\cdot\sigma)$, which has eigenphases $\pm\theta$, one computes $\|i\theta\,\hat{n}\cdot\sigma\| = 2\theta = \chi(g)$. The underlying manifold is thus the round 3-sphere of radius 2, with diameter 2π and total volume $\mathrm{Vol}(S^3) = 16\pi^2$; the *unit* round S^3 is the same manifold with every distance halved, and we write $d_1 = d/2$ and $\mathrm{Vol}_1(S^3) = 2\pi^2$ when that comparison is needed.

We fix the metric by this representation-theoretic rule, rather than by a choice of radius, because it is the rule under which the rate constant of Theorem 4.1 is comparable across groups: Paper 40 [23] uses the same normalisation for the whole compact connected semisimple class, so the present paper is exactly its rank-1 case. (Until 2026-09-03 this subsection declared the unit sphere while the moment $\gamma_{n_{\max}}$ of Lemma 3.4 was computed against χ; the two differ by the factor 2 above, and the bound was correspondingly stated with a unit-metric seminorm against a dual-Coxeter moment. Declaring the metric uniformly repairs the pairing in the main theorem: the distance d, the seminorms L and $L_{n_{\max}}$, the moment $\gamma_{n_{\max}}$ and the Monge–Kantorovich distances all scale together (a seminorm inversely, a distance moment directly), so the inequality is unchanged in content — and, since what was printed before paired a *unit*-metric seminorm with a dual-Coxeter moment, the uniform statement is a factor-2 *strengthening* of the printed bound rather than a restatement of equal force. Two scope limits: Lemmas 3.7–3.12 remain stated in the unit metric (Eq. (6) converts), and the panel *height* of §3.5 is scale-*invariant* rather than degree one — a difference of seminorms measured on that seminorm's own unit ball — so the panel comparison there is normalisation-dependent and is not part of the theorem.) Identify S^3 with the Lie group SU(2) via the standard double cover[2]. Let Σ denote the spinor bundle and $L^2(S^3, \Sigma)$ the spinor Hilbert space.

The Camporesi–Higuchi Dirac operator $D_{\mathrm{CH}} : \mathrm{Dom}(D_{\mathrm{CH}}) \subset L^2(S^3,\Sigma) \to L^2(S^3,\Sigma)$ is the Dirac operator of the *unit* round metric — we keep that normalisation of the operator throughout, so that (4) below is the familiar Camporesi–Higuchi spectrum; the Dirac operator of the dual-Coxeter metric declared above is $\frac{1}{2}D_{\mathrm{CH}}$. It acts on the spinor harmonics

$$D_{\mathrm{CH}}\,\psi_{n,l,m_j,\chi} \;=\; \chi\,(n+\tfrac{1}{2})\,\psi_{n,l,m_j,\chi}, \tag{4}$$

[2]We use this identification implicitly throughout. It is convenient because (i) the Peter–Weyl decomposition of $L^2(\mathrm{SU}(2))$ provides a natural orthonormal basis indexed by $j \in \frac{1}{2}\mathbb{N}_{\geq 0}$, (ii) the central conjugacy-class measure on SU(2) underlies the mass-concentration analysis of Lemma 3.4, and (iii) the spectrum of D_{CH} has the simplest closed form in this convention. We do not draw a structural distinction between S^3 and SU(2) in this paper.

with $n \in \mathbb{N}_{\geq 1}$, $l \in \{0, 1, \ldots, n-1\}$, $j = l + 1/2$, $m_j \in \{-j, -j+1, \ldots, j\}$, and chirality $\chi \in \{+1, -1\}$ [2, §2]. The degeneracy of the level n on each chirality sector is $n(n+1)$; the total degeneracy including both chiralities is $2n(n+1)$.

We then form the Camporesi–Higuchi spectral triple

$$\mathcal{T}_{S^3} \;=\; \big(C^\infty(S^3),\; L^2(S^3, \Sigma),\; D_{\mathrm{CH}}\big). \tag{5}$$

This is a unital, *odd*, real spectral triple of metric dimension 3 in the sense of Connes [3], with KO-dimension 3 (mod 8). *Parity corrected 2026-09-03:* the text read "even" here, which Connes' table forbids — the grading sign ε'' is defined only in even KO-dimension, so KO-3 carries no chirality, and on S^3 the label $\mathrm{diag}(+1, -1)$ is $\mathrm{sign}(D)$, which *commutes* with D and is therefore not a $\mathbb{Z}_2$ grading. Paper 32 already carries this reading ("KO-3 is the label of the odd continuum GV factor; S^3 carries no chirality") from the 2026-09-02 adjudication; that sweep did not reach this paper. The unitary real structure J given by charge conjugation and the Lipschitz seminorm enter as expected. Because D_{CH} is the unit-metric Dirac operator while d is the dual-Coxeter distance, the two differ by the same factor 2:

$$\|f\|_{\mathrm{Lip}} \;:=\; \|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \;=\; \|\nabla_1 f\|_{L^\infty}, \qquad L(f) \;=\; \mathrm{Lip}_d(f) \;=\; \tfrac{1}{2}\,\|f\|_{\mathrm{Lip}}, \tag{6}$$

so $\|\cdot\|_{\mathrm{Lip}}$ denotes throughout the *unit*-metric Dirac seminorm, while L — the dual-Coxeter Lipschitz seminorm, equivalently the translation seminorm of Definition 1.2 — is what metrizes every state space in the main theorem. Lemmas 3.7–3.12 below are stated in the unit metric (their gradients are the unit-metric ∇_1, as their polar-coordinate form shows), and the conversion between the two is Eq. (6); note that the good-kernel product $\mathrm{Lip}_d(f)\,\gamma_d$ is invariant under the metric scale, since a Lipschitz seminorm and a distance moment scale inversely, so an estimate of that shape is normalisation-independent *provided both factors are taken in the same metric.*

2.2 Truncations

Let $P_{n_{\max}} : L^2(S^3, \Sigma) \to \mathcal{H}_{n_{\max}}$ denote the orthogonal projection onto the finite-dimensional subspace spanned by $\{\psi_{n,l,m_j,\chi} : 1 \leq n \leq n_{\max}\}$. The truncated Hilbert space has dimension

$$\dim \mathcal{H}_{n_{\max}} \;=\; 2\sum_{n=1}^{n_{\max}} n(n+1) \;=\; \tfrac{2}{3}\,n_{\max}(n_{\max}+1)(n_{\max}+2). \tag{7}$$

The truncated Dirac operator is $D_{n_{\max}} = P_{n_{\max}} D_{\mathrm{CH}} P_{n_{\max}}$, which (since D_{CH} is diagonal in the basis above) acts as $\chi(n + 1/2)$ on each kept basis vector.

Following Connes–van Suijlekom [4, §3], the *truncated operator system* is

$$\mathcal{O}_{n_{\max}} \;=\; P_{n_{\max}}\, C^\infty(S^3)\, P_{n_{\max}} \;\subset\; B(\mathcal{H}_{n_{\max}}), \tag{8}$$

viewed as a self-adjoint subspace of the matrix algebra $M_{\dim \mathcal{H}_{n_{\max}}}(\mathbb{C})$. As emphasised in [4, §1], this is *not* a C^*-subalgebra of $M_{\dim \mathcal{H}_{n_{\max}}}(\mathbb{C})$: the full C^*-algebra it generates is the full matrix algebra (the "C^*-envelope"), but the operator system $\mathcal{O}_{n_{\max}}$ itself encodes finer structure, including the algebra-side analog of the Lipschitz seminorm. The truncated metric spectral triple is then

$$\mathcal{T}_{n_{\max}} \;=\; (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}}). \tag{9}$$

2.3 Peter–Weyl basis and the Avery 3-Y integral

The Peter–Weyl decomposition of $L^2(\mathrm{SU}(2))$ gives an orthonormal basis $\{D^j_{m,m'} : j \in \tfrac{1}{2}\mathbb{N}_{\geq 0}, |m|, |m'| \leq j\}$ of Wigner D-matrix elements. Setting $n = 2j + 1$ identifies the Peter–Weyl level with the Fock-shell index used above. We work in the *Wen–Avery hyperspherical harmonic* basis

$$Y^{(3)}_{NLM}(\chi, \theta, \phi) \;=\; \mathcal{N}_{NL}\, \sin^L \chi \cdot C^{L+1}_{N-L-1}(\cos\chi)\, Y_{LM}(\theta, \phi), \tag{10}$$

which is the standard SO(4)-adapted basis of $L^2(S^3)$ in the form given by Wen and Avery [1] and used systematically in the molecular orbital literature; here $\mathcal{N}_{NL} = \sqrt{2^{2L+1}N(N-L-1)!(L!)^2/(N+L)!}/\sqrt{\pi}$ is the standard normalisation, C^{L+1}_{N-L-1} is the Gegenbauer polynomial, and Y_{LM} are the standard spherical harmonics on S^2. The Laplace–Beltrami spectrum on S^3 is then $-(N^2-1)$ on the isotypic component spanned by $\{Y^{(3)}_{NLM}\}_{L,M}$.

For $f = \sum_{N\geq 1}\sum_{L=0}^{N-1}\sum_{|M|\leq L} c_{NLM}\, Y^{(3)}_{NLM}$, the multiplier matrix $M_f := P_{n_{\max}} f P_{n_{\max}}$ acts on $\mathcal{H}_{n_{\max}}$ with entries

$$(M_f)_{(n',l',m'_j),(n,l,m_j)} \;=\; \sum_{N,L,M} c_{NLM}\, \mathcal{G}^{(3)}_{NLM;nlm_j;n'l'm'_j}, \tag{11}$$

where the Avery 3-Y integral $\mathcal{G}^{(3)}$ is the Clebsch–Gordan- weighted sum of Gegenbauer triple integral and S^2 Gaunt integral factors[3].

The selection rule $|n-n'|+1 \leq N \leq n+n'-1$ enforces a sparse structure on M_f that we use crucially in Lemma 3.7.

3 The five lemmas

3.1 Lemma L1′: chirality-doubled operator system

The chirality-doubled spinor bundle $\Sigma = \Sigma_+ \oplus \Sigma_-$ gives the truncated Hilbert space the structure $\mathcal{H}_{n_{\max}} = \mathcal{H}^+_{n_{\max}} \oplus \mathcal{H}^-_{n_{\max}}$, with each chirality block carrying a Weyl-sector multiplier matrix M_f^{Weyl} acting identically on the two blocks. The multiplier M_f is therefore block-diagonal in chirality:

$$M_f \;=\; M_f^{\text{Weyl}} \oplus M_f^{\text{Weyl}} \quad \text{on} \quad \mathcal{H}^+_{n_{\max}} \oplus \mathcal{H}^-_{n_{\max}}. \tag{12}$$

Lemma 3.1 (L1′: chirality-doubled operator system substrate; **[SYMBOLIC + MEASURED]**). *The operator system $\mathcal{O}_{n_{\max}}$ inherits from (12) a chirality block-diagonal structure. The truncated Connes distance*

$$d_{\mathcal{T}_{n_{\max}}}(\varphi,\psi) \;=\; \sup\big\{\, |\varphi(M)-\psi(M)| \;:\; M \in \mathcal{O}_{n_{\max}},\, \|[D_{n_{\max}},M]\|_{\text{op}} \leq 1 \,\big\} \tag{13}$$

is finite on every cross-shell pair of pure node-evaluation states when (12) is replaced by the chirality-mixing offdiagonal Camporesi–Higuchi extension $D^{\text{offdiag}}_{n_{\max}}$ *obtained by adding the standard $E1$ ladder couplings between $(n,l,m_j,+)$ and $(n+1,l\pm 1,m_j,-)$.*

Proof. The block-diagonal structure of M_f in (12) follows directly from the chirality-blindness of scalar multipliers. To verify finiteness of the Connes distance on cross-shell pairs, we observe that a state pair $(\varphi_{n,l,m_j},\psi_{n',l',m'_j})$ with $n \neq n'$ has finite truncated Connes distance under the offdiag extension because the $E1$ coupling supplies a non-trivial structural multiplier that bounds the supremum in (13). **[PANEL-VERIFIED]** Numerical verification at $n_{\max} \in \{2,3\}$ shows every cross-shell pair has finite distance, with the four-point Pearson correlation between distance and Fock-graph hop distance taking the value -0.2501 at $n_{\max}=3$; no frozen test in the repository backs this correlation value.[4]

[3] In SO(4) the relevant 3-Y integral on S^3 factors as a Gegenbauer triple integral times a standard 3-Y integral on S^2, with the SO(4) selection rule $|n-n'|+1 \leq N \leq n+n'-1$ enforcing the truncation $|n-n'|<N$ on the radial channel and the standard S^2 Gaunt selection rule on the angular channel. This double-selection structure is essential to Lemma 3.7 and is treated as canonical infrastructure here. Closed-form values used in internal numerical verifications appear in the supporting code module `geovac/so4_three_y_integral.py` (R3.1, May 2026).

[4] The numerical correlation value is recorded in the internal computation log; the structural finiteness is the load-bearing input for Lemma 3.12, and is robust under operator-norm perturbations of D that preserve the SU(2) selection rules.

The role of the offdiag extension is purely SDP-bounding: it ensures that the operator-norm constraint $\|[D_{n_{\max}}, M]\|_{\mathrm{op}} \leq 1$ is non-degenerate on the full operator system. The truthful D_{CH} (without offdiag couplings) is what underlies the spectral-action structure of $\mathcal{T}_{S^3}$ and is what we use in Lemmas 3.7–3.10. The two operators are related by a bounded compact perturbation, and the state-space GH bound of Theorem 1.1 is robust under this choice. □

Remark 3.2. The offdiag extension does *not* satisfy $\{\gamma_5, D^{\mathrm{offdiag}}_{n_{\max}}\} = 0$, whereas the truthful D_{CH} does. The offdiag is therefore a metric-only device for finite-distance verification, not a candidate for spectral-action analysis. The truthful D_{CH} is what enters (4) and what we use throughout the state-space GH convergence proof.

3.2 Lemma L2: central spectral Fejér kernel on SU(2)

We construct the central spectral Fejér kernel $K_{n_{\max}} \in C(\mathrm{SU}(2))$ that plays the role of the abelian-compact-group Fejér kernel in Connes–van Suijlekom [4, §3] but with the natural-coefficient Plancherel weights forced by the Peter–Weyl bijection $n = 2j + 1$.

Definition 3.3 (Central spectral Fejér kernel)**.** For $n_{\max} \geq 1$, define

$$K_{n_{\max}}(g) := \frac{1}{Z_{n_{\max}}} \left| \sum_{j \leq j_{\max}} \sqrt{2j+1}\, \chi_j(g) \right|^2, \qquad Z_{n_{\max}} = \frac{n_{\max}(n_{\max}+1)}{2}, \tag{14}$$

where χ_j is the trace character of the $(2j+1)$-dimensional irreducible representation of SU(2) and $j_{\max}$ is the largest half-integer with $2j_{\max} + 1 \leq n_{\max}$.

The kernel is positive on SU(2) (manifestly), central (depends only on the conjugacy class of g), and unit-mass under the standard SU(2) Haar measure dg. Identifying conjugacy classes with the rotation angle $\chi \in [0, 2\pi]$ via $g = \exp(i(\chi/2)\,\hat{n}\cdot\sigma)$ on the maximal torus (eigenphases $\pm\chi/2$; unit-S^3 geodesic distance $\chi/2$; corrected 2026-09-03 from $\chi \in [0, \pi]$ with $g = \exp(i\chi\hat{n}\cdot\sigma)$, which is the geodesic angle and does not match the characters below), the conjugacy-class measure is $d\mu(\chi) = (1/\pi)\sin^2(\chi/2)\, d\chi$.

Lemma 3.4 (L2: kernel construction and quantitative rate)**.** *The central spectral Fejér kernel $K_{n_{\max}}$ has the following properties (the closed-form $\gamma_{n_{\max}}$ rate moment is verified by* `tests/test_central_fejer` *and the $4/\pi$ asymptotic constant by* `tests/test_trunk_qa_fejer_4_over_pi.py`*; $2/\pi$ is the same constant on the unit S^3 and the circle's constant, not a decoy):*

(a) $K_{n_{\max}}(g) \geq 0$ for all g, and $\int_{\mathrm{SU}(2)} K_{n_{\max}}(g)\, dg = 1$.

(b) (Mass distribution and multiplier symbol.) The kernel's Plancherel mass distribution *is $m_{n_{\max}}(j) = (2j+1)/Z_{n_{\max}}$ on $j \leq j_{\max}$ (the squared Plancherel weights of the amplitude $\sum_j \sqrt{2j+1}\,\chi_j$), a probability distribution with maximum $2/(n_{\max}+1)$ at $j_{\max}$. The convolution operator $T_{K_{n_{\max}}} : f \mapsto K_{n_{\max}} * f$ acts on the j'-isotypic component by the* multiplier symbol $\sigma_{n_{\max}}(j') = c_{j'}/((2j'+1)Z_{n_{\max}})$ *with $c_{j'} = \sum_{j_1, j_2 \leq j_{\max},\, \triangle(j_1,j_2,j')} \sqrt{(2j_1+1)(2j_2+1)}$, where $\triangle$ is the Clebsch–Gordan condition* with unit steps *($|j_1 - j_2| \leq j' \leq j_1 + j_2$ and $j_1 + j_2 + j' \in \mathbb{Z}$): $\sigma_{n_{\max}}(0) = 1$, $0 \leq \sigma_{n_{\max}} \leq 1$, support $j' \leq 2j_{\max}$.* ***[MEASURED]*** *At $n_{\max} = 2$: $\sigma = (1, \sqrt{2}/3, 2/9)$ at $j' = (0, \frac{1}{2}, 1)$ (bit-exact against independent Weyl-integration quadrature).*

(c) (Cb-norm.) $T_{K_{n_{\max}}}$ is unital completely positive (convolution against a probability density), hence $\|T_{K_{n_{\max}}}\|_{\mathrm{cb}} = 1$. (The mass-distribution maximum $2/(n_{\max}+1)$ of (b) is not *the cb-norm of any map used in this paper.)*

(d) (Mass-concentration moment.) Define

$$\gamma_{n_{\max}} := \int_{\mathrm{SU}(2)} K_{n_{\max}}(g)\,\chi(g)\,dg, \tag{15}$$

where $\chi(g) \in [0, 2\pi]$ is the class (rotation) angle of g, which by §2.1 is the dual-Coxeter geodesic distance $d(e, g)$ ($g = \cos\theta + i \sin\theta\, \hat{n} \cdot \sigma$ has eigenphases $\pm\theta$ and $d(e, g) = 2\theta = \chi$; the characters below are $\sin((2j{+}1)\chi/2)/\sin(\chi/2)$). So $\gamma_{n_{\max}}$ is the mass-concentration moment against the paper's own metric, and every constant below is a dual-Coxeter constant; the corresponding unit-S^3 moment is exactly $\gamma_{n_{\max}}/2$ at every $n_{\max}$ (`tests/test_p38_metric_conve`). $K_1 \equiv 1$ makes $\gamma_1 = \pi$, exactly twice the unit-S^3 Haar-mean distance $\pi/2$). Correction (2026-09-03): *this definition previously wrote "d_{round}" while §2.1 declared the unit sphere, so the moment and the seminorm were stated in different metrics; both are now dual-Coxeter. Then:*

(d.i) ***[INTERNAL THEOREM]*** *$\gamma_{n_{\max}}$ admits the closed-form sum-rule*

$$\gamma_{n_{\max}} = \pi - \frac{4T_{n_{\max}}}{\pi\, Z_{n_{\max}}}, \qquad T_{n_{\max}} := \sum_{\substack{1 \le k_1, k_2 \le n_{\max} \\ k_1 + k_2 \text{ odd}}} \sqrt{k_1 k_2}\left[\frac{1}{(k_1 - k_2)^2} - \frac{1}{(k_1 + k_2)^2}\right]. \tag{16}$$

(d.ii) (Asymptotic rate.) ***[INTERNAL THEOREM + MEASURED]*** $\lim_{n_{\max}\to\infty} \dfrac{n_{\max}\,\gamma_{n_{\max}}}{\log n_{\max}} = \dfrac{4}{\pi}$.

(d.iii) (Uniform bound.) ***[PANEL-VERIFIED]*** *$\gamma_{n_{\max}} \le 6 \log n_{\max}/n_{\max}$ for all $n_{\max} \ge 2$ (established through a monotonicity statement checked on a finite range of $n_{\max}$, not proved for all $n_{\max}$).*

Proof sketch. Property (a) is a standard Peter–Weyl computation. For (b), the mass distribution is immediate from the amplitude weights; the multiplier symbol follows by expanding $|h|^2$ with the SU(2) Clebsch–Gordan rule $\chi_{j_1}\chi_{j_2} = \sum_{J=|j_1-j_2|}^{j_1+j_2} \chi_J$ (unit steps in J, so $j_1 + j_2 + J \in \mathbb{Z}$) and reading off the convolution eigenvalue $c_{j'}/((2j'+1)Z)$ on each block; unitality at $j' = 0$ is $c_0 = \sum_j (2j+1) = Z$, and the $n_{\max} = 2$ values are verified bit-exactly against independent Weyl-integration quadrature (supporting computation in the project repository, `debug/p45_adversarial_L2_memo.md`). Property (c): a unital positive map on a commutative domain is completely positive, and a UCP map has cb-norm exactly 1; on the truncated side the induced central multiplier acts block-scalar on $\bigoplus_\pi M_{d_\pi}(\mathbb{C})$ with cb-norm $\sup_\pi |\sigma(\pi)| = 1$ — an elementary statement requiring no amenability input.

For (d), the closed form (d.i) follows from rewriting the kernel using $\sin a \sin b = \frac{1}{2}(\cos(a - b) - \cos(a + b))$ and applying the elementary integral $\int_0^{2\pi} \chi \cos(m\chi/2) d\chi = -8/m^2$ for odd $m \neq 0$, $2\pi^2$ for $m = 0$, and 0 for non-zero even m. Substituting and separating diagonal from off-diagonal contributions gives (16).

For (d.ii), expand $\sqrt{k_1 k_2} = \sqrt{a(a + d)} \approx a + d/2 + O(d^2/a)$ for $k_2 = k_1 + d$, d odd, $a \gg d$. The leading-order sum is $\sum_{d\,\mathrm{odd}} 1/d^2 \cdot \sum_a a \to (\pi^2/8) \cdot n^2/2$, giving $T_n/Z_n \to \pi^2/4$ and $\gamma_n \to 0$. The sub-leading $\log n/n$ correction is the net of two logarithmic pieces — the triangle truncation of the odd-d sum (coefficient 2) and the $d/2$ term of $\sqrt{a(a + d)}$ (coefficient -1) — computed in Appendix A, Step 3:

$$\frac{T_n}{Z_n} = \frac{\pi^2}{4} - \frac{\log n}{n} + O(1/n),$$

which gives $\gamma_n = (4/\pi) \log n/n + O(1/n)$.

For (d.iii), **[PANEL-VERIFIED]** the function $n \mapsto n\gamma_n/\log n$ is monotonically decreasing for $n \ge 3$ (verified on the sample $n \in \{3, 4, 5, 7, 10, 15, 20, 30, 50, 100, 200, 500, 1000\}$ by the

frozen test `tests/test_central_fejer_su2.py`; no proof of monotonicity for all n is given), with maximum at $n = 2$ equal to $2\gamma_2/\log 2 = 2(\pi - 64\sqrt{2}/(27\pi))/\log 2 \approx 5.986$. The uniform bound follows. □

Remark 3.5 (Connection to the circle Fejér estimate)**.** The asymptotic rate $\gamma_n \sim (4/\pi)\log n/n$ is the non-abelian analog of the classical Fejér-on-the-circle estimate

$$\frac{1}{2\pi}\int_{\mathbb{T}^1} F_n(\theta)\,|\theta|\,d\theta \;\sim\; \frac{2}{\pi}\,\frac{\log n}{n}, \qquad F_n(\theta) = \frac{1}{n}\,\frac{\sin^2(n\theta/2)}{\sin^2(\theta/2)}, \tag{17}$$

for the probability-normalised Fejér kernel ($\frac{1}{2\pi}\int_{\mathbb{T}^1} F_n = 1$; Zygmund [29] Vol. I, Ch. III). The circle moment has the exact closed form $\frac{1}{2\pi}\int_{\mathbb{T}^1} F_n|\theta| = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k\ \mathrm{odd},\,k<n}(1-k/n)/k^2$ (from $\int_0^\pi \theta\cos k\theta\,d\theta = ((-1)^k-1)/k^2$), whose $\log n/n$ coefficient is $2/\pi$ (the $\frac{1}{n}\sum_{k\ \mathrm{odd}} 1/k \sim \frac{1}{2n}\log n$ term). **[SYMBOLIC + MEASURED]** The circle constant is $2/\pi$, not $4/\pi$: the doubling estimator $(2n\,m_{2n} - n\,m_n)/\log 2$ on the closed form gives 0.63662 at $n = 400$ ($2/\pi = 0.63662$), the SU(2) doubling estimator of Lemma 3.4(d.ii), $a_n = (2n\gamma_{2n} - n\gamma_n)/\log 2$, reads 1.2785 at $n = 800$ and 1.2761 at $n = 1600$ (i.e. using γ_{1600} and γ_{3200}; three digits of $4/\pi = 1.2732$, approaching from above with the residual shrinking by $\sim$1.85× per doubling; Appendix A), and the ratio of the two estimators is 2.008 at $n = 800$ (`tests/test_trunk_qa_fejer_4_over_pi.py`: `test_circle_fejer_closed_form_matches_quadrature`, `test_circle_fejer_constant_is_2_over_pi`, `test_su2_constant_is_twice_circle`). An earlier version of this remark stated the circle constant as $4/\pi$, "the same on both sides"; that statement is withdrawn (2026-09-02 certification run). **[OBSERVATION]** In the dual-Coxeter normalisation the SU(2) constant is twice the circle constant; in the unit-S^3 normalisation (geodesic distance, the circle's own convention) the two coincide at $2/\pi$ (circle: $2\mathrm{Vol}(S^0)/\mathrm{Vol}(S^1) = 4/(2\pi) = 2/\pi$; SU(2): $\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = 4\pi/(2\pi^2) = 2/\pi$ — the same number, reached with a factor 2 on the circle side and without it on the SU(2) side, so no common "2Vol(base)/Vol(group)" form is claimed), the ratio of the unit-metric SU(2) estimator to the circle's being 1.004 at $n = 800$ (2026-09-03 correction: the factor 2 reported before is the metric scale, not a group-theoretic signature). No derivation of the SU(2) constant from the circle constant is given here; the $\sin^2(\chi/2)$ conjugacy-class weight and the $\sqrt{2j+1}$ Plancherel weight of Definition 3.3 together change both the kernel and the constant, while preserving the leading $\log n/n$ behaviour that the Cesàro-1 (Fejér) first moment already has on the circle. The one-log slowdown quoted in the abstract is relative to the Leimbach–van Suijlekom torus rate $O(1/n_{\max})$ [17], not relative to the circle Fejér first moment.

Remark 3.6 (The volume ratio behind the constant)**. [OBSERVATION]** In the dual-Coxeter normalisation the asymptotic constant coincides numerically with

$$\frac{4}{\pi} \;=\; \frac{\mathrm{Vol}(S^2)}{\pi^2} \;=\; \frac{4\pi}{\pi^2}.$$

Which volume ratio (2026-09-03). The constant is a genuine quotient of sphere volumes — but not the one its usual name suggests. For unit-radius spheres $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 4\pi/2\pi^2 = 2/\pi$ exactly, and the dual-Coxeter constant is twice that. It is *not* the Hopf fibration's base-to-total ratio: the Hopf map is a Riemannian submersion $S^3(r) \to S^2(r/2)$, so that ratio is $1/(2\pi)$ on the unit sphere and $1/(4\pi)$ in the dual-Coxeter metric. The *volume content* of the constant — what places it in the $k = 0$ (M1) slot of the master Mellin engine and in $\mathbb{Q}[\pi, \pi^{-1}]$ — is therefore intact; only the "Hopf base" label is a misnomer, that base carrying half the radius of the sphere whose volume appears. Read as the unit-sphere quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ rather than as a Hopf base-to-total ratio, the constant still carries pure volume content, and is one of the natural transcendentals on SU(2) that survives at the leading-order rate of any Cesaro-2-style averaging procedure.

3.3 Lemma L3: Lipschitz comparison with $C_3 = 1$

The next lemma is the Lipschitz comparison estimate that controls the operator norm of the commutator $[D_{\mathrm{CH}}, M_f]$ in terms of the classical Lipschitz norm of f on the round S^3.

Lemma 3.7 (L3: Lipschitz comparison, truthful D_{CH} on S^3; **[PANEL-VERIFIED]**). *For every $f \in C^\infty(S^3)$ and every $n_{\max} \geq 1$,*

$$\|[D_{\mathrm{CH}},\, M_f]\|_{\mathrm{op}} \;\leq\; C_3 \cdot \|\nabla f\|_{L^\infty(S^3)}, \qquad C_3 = 1, \tag{18}$$

where the round-S^3 Lipschitz norm $\|\nabla f\|_{L^\infty}^2 = (\partial_\chi f)^2 + \sin^{-2}\chi\,(\partial_\theta f)^2 + \sin^{-2}\chi \sin^{-2}\theta\,(\partial_\phi f)^2$ is taken in the standard polar parametrisation.

Frozen falsifier: `tests/test_r25_l3_lipschitz_bound.py` (`test_eq_l3_main_inequality`, `test_l3_C3_unifo`

Proof. The truthful D_{CH} is diagonal in the basis $\{\psi_{n,l,m_j,\chi}\}$ with eigenvalue $\chi(n+1/2)$, so

$$\big[D_{\mathrm{CH}}, M_f\big]_{(n',l',m_j',\chi'),\,(n,l,m_j,\chi)} \;=\; \delta_{\chi,\chi'}\,\chi\,(n-n')\,(M_f^{\mathrm{Weyl}})_{(n',l',m_j'),\,(n,l,m_j)}. \tag{19}$$

For each isotypic component of f at hyperspherical level N, the SO(4) selection rule $|n-n'|+1 \leq N \leq n+n'-1$ on the multiplier matrix M_{NLM} forces $|n-n'| \leq N-1$, so the shell-difference factor in (19) is bounded entrywise by $N-1$:

$$\|[D_{\mathrm{CH}},\, M_{NLM}]\|_{\mathrm{op}} \;\leq\; (N-1)\cdot\|M_{NLM}\|_{\mathrm{op}}. \tag{20}$$

The Laplace–Beltrami spectrum on S^3 at level N is $-(N^2-1)$, so the round-S^3 Lipschitz norm of a unit-normalised harmonic $Y^{(3)}_{NLM}$ scales as

$$\|\nabla Y^{(3)}_{NLM}\|_{L^\infty} \;\sim\; \sqrt{N^2-1}\;\cdot\;\|M_{NLM}\|_{\mathrm{rel}}, \tag{21}$$

where **[PANEL-VERIFIED]** the relative-multiplier-norm prefactor is bounded above by 1 on the unit-normalised panel[5].

Combining (20) with (21), the per-harmonic Lipschitz comparison ratio satisfies

$$\frac{\|[D_{\mathrm{CH}}, M_{NLM}]\|_{\mathrm{op}}}{\|\nabla Y^{(3)}_{NLM}\|_{L^\infty}} \;\leq\; \frac{N-1}{\sqrt{N^2-1}} \;=\; \sqrt{\frac{N-1}{N+1}} \;<\; 1, \tag{22}$$

with the right-hand side approaching 1 from below as $N \to \infty$. By the linear triangle inequality and the dual triangle inequality on the Lipschitz norm, the per-harmonic bound (22) implies the global bound

$$\frac{\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}}}{\|\nabla f\|_{L^\infty}} \;\leq\; \sup_N \sqrt{\frac{N-1}{N+1}} \;\leq\; 1$$

for any finite combination $f = \sum c_{NLM} Y^{(3)}_{NLM}$, and the result extends to $C^\infty(S^3)$ by density. □

Remark 3.8. [PANEL-VERIFIED] The bound $C_3 = 1$ is approached as $N \to \infty$ and is strictly below 1 at every finite cutoff $n_{\max}$; we do not claim it is attained. Specifically, $C_3(n_{\max}) \leq \sup_{N \leq n_{\max}} \sqrt{(N-1)/(N+1)} = \sqrt{(n_{\max}-1)/(n_{\max}+1)}$ on the sub-envelope $N \leq n_{\max}$ — the multiplier substrate in fact reaches $N \leq 2n_{\max}-1$ (Definition 3.9), so the per-cutoff values listed next bound the sub-envelope only. That value is 0 at $n_{\max} = 1$, $1/\sqrt{3} \approx 0.577$ at $n_{\max} = 2$,

[5] This is verified numerically on the test panel $\{Y^{(3)}_{NLM} : 2 \leq N \leq 4\}$ in supporting computations only; a general proof of the inequality $\|\nabla Y\|_{L^\infty} \leq \sqrt{N^2-1}\,\|M_{NLM}\|_{\mathrm{rel}}$ is open (the natural Sobolev route needs H^s with $s > 3/2$ on S^3, not H^1, so the low-harmonic smoothness must enter quantitatively). Because $C_3 \leq 1$ is derived through (21), the $C_3 \leq 1$ statement is panel-verified on the stated test set rather than proved in general; the unconditional main theorem does not use Lemma L3 and is unaffected.

$1/\sqrt{2} \approx 0.707$ at $n_{\max} = 3$, $\sqrt{3/5} \approx 0.775$ at $n_{\max} = 4$, and so on. Taking the supremum over the full substrate range instead would give $\sqrt{1 - 1/n_{\max}}$. *That expression must not be substituted here without a measurement.* It is textually identical to a constant retired in 2026-06 (Papers 45/46): the *operator-norm*-normalised envelope $\|[D_{\mathrm{GV}}, M]\|_{\mathrm{op}}/\|M\|_{\mathrm{op}} \leq \sqrt{1 - 1/n_{\max}}$, which is false on the operator system. The bound (22) above is normalised by $\|\nabla Y\|_{L^\infty}$, not by $\|M\|_{\mathrm{op}}$, and is a different statement that survives that retraction. The correct gradient-normalised per-cutoff envelope is an open measurement, not an inference from the substrate range; the load-bearing conclusion $C_3 \leq 1$ is unaffected either way.

3.4 Lemma L4: Berezin reconstruction

Definition 3.9 (Berezin map, convolution form)**.** For $n_{\max} \geq 1$ and $f \in C^\infty(S^3)$, define

$$B_{n_{\max}}(f) \;:=\; P_{n_{\max}}\, M_{K_{n_{\max}} * f}\, P_{n_{\max}} \;=\; \sum_{N \leq 2n_{\max}-1} \sum_{L=0}^{N-1} \sum_{|M| \leq L} \sigma_{n_{\max}}(N)\, c_{NLM}\, M_{NLM}, \qquad (23)$$

where $\sigma_{n_{\max}}(N)$ is the multiplier symbol of Lemma 3.4(b) under the bijection $n = 2j + 1$ and $f = \sum c_{NLM} Y^{(3)}_{NLM}$. (A sum form weighted by the mass distribution $N/Z_{n_{\max}}$ in place of $\sigma_{n_{\max}}(N)$ would not be unital — $B(\mathbf{1}) = Z^{-1} I$ — and would fail the approximate-identity property maximally at $f = \mathbf{1}$. Note the image reaches multiplier labels up to the achievable envelope $N \leq 2n_{\max} - 1$.)

Lemma 3.10 (L4: properties of the Berezin reconstruction)**.** *The map $B_{n_{\max}} : C^\infty(S^3) \to \mathcal{O}_{n_{\max}}$ has the following four properties:*

(a) (Positivity.) If $f \geq 0$ on S^3, then $B_{n_{\max}}(f)$ is positive semi-definite.

(b) (Contractivity.) $\|B_{n_{\max}}(f)\|_{\mathrm{op}} \leq \|f\|_{L^\infty}$.

(c) (Approximate identity.) $\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}} \leq \gamma_{n_{\max}} \cdot \|\nabla f\|_{L^\infty}$.

*(d) (Compatibility with L3.) **[PANEL-VERIFIED]** $\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \leq \|\nabla f\|_{L^\infty}$ (inherits the panel scope of Lemma 3.7 through $C_3 = 1$).*

Proof. **(a)** Equivalent to the convolution form $B_{n_{\max}}(f) = P_{n_{\max}}(K_{n_{\max}} * f) P_{n_{\max}}$: since $K_{n_{\max}} \geq 0$ pointwise (Lemma 3.4(a)) and the convolution of two non-negative functions on a compact group is non-negative, $K_{n_{\max}} * f \geq 0$. Compression by an orthogonal projection preserves positivity: for any $\psi \in \mathcal{H}_{n_{\max}}$,

$$\langle \psi | P_{n_{\max}} M_g P_{n_{\max}} | \psi \rangle \;=\; \int_{S^3} g(\omega)\, |P_{n_{\max}} \psi(\omega)|^2 \, d\Omega \;\geq\; 0.$$

(b) By Young's inequality, $\|K_{n_{\max}} * f\|_{L^\infty} \leq \|K_{n_{\max}}\|_{L^1} \|f\|_{L^\infty} = \|f\|_{L^\infty}$. Compression contracts operator norms, so $\|B_{n_{\max}}(f)\|_{\mathrm{op}} \leq \|f\|_{L^\infty}$.

(c) The standard Lipschitz/good-kernel estimate gives

$$\|K_{n_{\max}} * f - f\|_{L^\infty} \;\leq\; \int_{\mathrm{SU}(2)} K_{n_{\max}}(g)\, |f(g \cdot x) - f(x)|\, dg \;\leq\; \gamma_{n_{\max}}\, \|\nabla f\|_{L^\infty},$$

which is the SU(2) version of the standard approximate-identity estimate (Stein–Weiss [28], Ch. I). Compression then gives $\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}} \leq \gamma_{n_{\max}} \|\nabla f\|_{L^\infty}$.

(d) The cleanest argument goes via the convolution form $B_{n_{\max}}(f) = P_{n_{\max}}\, M_{K_{n_{\max}} * f}\, P_{n_{\max}}$ used in parts (a) and (c). Let $g := K_{n_{\max}} * f \in C^\infty(S^3)$. Since $K_{n_{\max}} \geq 0$ and $\|K_{n_{\max}}\|_{L^1} = 1$ by Lemma 3.4(a), Young's inequality for the gradient

$$\|\nabla g\|_{L^\infty} \;=\; \|\nabla (K_{n_{\max}} * f)\|_{L^\infty} \;=\; \|K_{n_{\max}} * \nabla f\|_{L^\infty} \;\leq\; \|K_{n_{\max}}\|_{L^1}\, \|\nabla f\|_{L^\infty} \;=\; \|\nabla f\|_{L^\infty} \qquad (24)$$

gives a smoothed function g whose Lipschitz norm is no larger than that of f (left-translation by $K_{n_{\max}}$ commutes with the gradient on the bi-invariant metric and convolution with a probability density does not increase the Lipschitz norm). Now apply Lemma 3.7 to g in the truncated multiplier representation: $B_{n_{\max}}(f) = M_g$ in $\mathcal{O}_{n_{\max}}$, so

$$\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \;=\; \|[D_{\mathrm{CH}}, M_g]\|_{\mathrm{op}} \;\le\; C_3\,\|\nabla g\|_{L^\infty} \;\le\; C_3\,\|\nabla f\|_{L^\infty} \;=\; \|\nabla f\|_{L^\infty},$$

where the final equality uses $C_3 = 1$ from Lemma 3.7. □

Remark 3.11 (Non-Kähler analog)**.** The Berezin reconstruction (23) differs from the standard Hawkins [5] construction on Kähler manifolds, which proceeds via the holomorphic Toeplitz form on a Hardy space. Since $S^3 = \mathrm{SU}(2)$ admits no compatible almost-complex structure (it is parallelisable but not Kähler), the natural analog is the central-spectral-Fejér-kernel form, which is the abelian-compact-group version of the same idea applied to the conjugacy-class measure on SU(2). The Kähler analog on $S^2 = \mathrm{SU}(2)/U(1)$ via Berezin–Toeplitz on the Hopf base has been worked out independently (see [9] and references therein) and is consistent with the present construction in the appropriate fibration limit.

Panel note (2026-09-03). The reference implementation (`geovac/gh_convergence.py`) reports reach and height on a fixed test panel of low-degree harmonics. Those panel numbers do *not* verify this lemma: the Lipschitz-normalised panel height rises with the cutoff while $\gamma_{n_{\max}}$ falls, and the two cross between $n_{\max} = 6$ and $n_{\max} = 7$ (measured 2026-09-03):

$n_{\max}$	2	3	4	5	6	7
$\gamma_{n_{\max}}$	2.075	1.610	1.322	1.130	0.990	0.883
height/$\|f\|_{\mathrm{Lip}}$	0.667	0.808	0.878	0.913	0.937	0.952
margin	+1.408	+0.803	+0.445	+0.217	+0.053	−0.069

Normalised as printed, the panel entries rescale into the unit-Lipschitz ball, so they are a subset of the set quantified over and a lower bound on the supremum: an entry exceeding $\gamma_{n_{\max}}$ contradicts the withdrawn bound. The analytic witness of (28) settles the question independently of the panel. The proof below does not use the panel; identifying the panel-side quantity is an open check.

3.5 Lemma L5: state-space Gromov–Hausdorff assembly

A remark on framework before assembling. In Latrémolière's quantum Gromov–Hausdorff propinquity for metric spectral triples [15, 16], the distance is an infimum over tunnels carried by quantum metric spaces with quantum isometries, under hypotheses (Leibniz Lip-norms on C*-algebras; kernel condition) not established for the truncated operator system; a pair of UCP maps is not a tunnel there. The distance the pair $(B_{n_{\max}}, P_{n_{\max}})$ certifies is van Suijlekom's state-space Gromov–Hausdorff distance [8], via two-directional approximation data in the Leimbach–van Suijlekom pattern [17]. We write

$$\Lambda(\mathcal{T}_1, \mathcal{T}_2) \;:=\; \mathrm{d}_{\mathrm{GH}}^{\mathrm{vS}}\big(S(\mathcal{O}_1),\, S(\mathcal{O}_2)\big) \tag{25}$$

for this distance (between state spaces with their respective Monge–Kantorovich metrics), and retain the constituent names reach_B, reach_P, height_B, height_P for the four approximation estimates the pair supplies. The truncated-side Monge–Kantorovich metric is non-degenerate unconditionally under the translation-seminorm metrization (Proposition 1.5); the distance bound itself is Theorem 1.7, and the pair below supplies the explicit forward-reconstruction refinement.

Lemma 3.12 (L5: Berezin-pair refinement of the forward direction; **[REFUTED AS STATED — withdrawn 2026-09-03]**). *Within the framework of § 1.1 (kernel condition Proposition 1.5;*

distance bound Theorem 1.7), the pair $(B_{n_{\max}}, P_{n_{\max}})$ *with* $B_{n_{\max}}$ *from Definition 3.9 and* $P_{n_{\max}}$ *the standard spectral projection yields the bound*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\leq\; C_3 \cdot \gamma_{n_{\max}}, \tag{26}$$

with $C_3 = 1$ *(Lemma 3.7) and* $\gamma_{n_{\max}}$ *from Lemma 3.4(d).*

Proof. The four constituents a Latrémolière tunnel would require are:

1. $\mathrm{reach}_B := \sup_{\|f\|_{\mathrm{Lip}} \leq 1} \|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}}$,
2. reach_P, the dual reach via the partial inverse,
3. height_B, the Lipschitz-distortion envelope of $B_{n_{\max}}$ (defined precisely below),
4. $\mathrm{height}_P = 0$ for the orthogonal projection *(withdrawn: false by the same finite-band witness as* height_B*, see below).*

Reach. By L4(c), $\mathrm{reach}_B \leq \gamma_{n_{\max}} \cdot \|\nabla f\|_{L^\infty} \leq \gamma_{n_{\max}}$ on the unit Lipschitz ball $\{\|f\|_{\mathrm{Lip}} \leq 1\}$, where **[PANEL-VERIFIED]** we have used L3 to convert $\|\nabla f\|_{L^\infty} \leq \|f\|_{\mathrm{Lip}}$ with constant $C_3 = 1$, so this route — unlike Theorem 1.7 — carries Lemma 3.7's panel scope. **(Dissolved.)** The dual estimate is no longer needed: Theorem 1.7 obtains the distance bound via the lifted-state map, whose almost-inverse defect is the conjugation average Φ at the same moment $\gamma_{n_{\max}}$ (Remark 1.8) — no partial inverse of $B_{n_{\max}}$, and no transference estimate, enters anywhere. (For the record: a partial inverse of $B_{n_{\max}}$ on the full support window would be governed by the symbol *minimum* and grow like $O(n_{\max}^2)$, which is why no forward bound could supply it.) The pair $(B_{n_{\max}}, P_{n_{\max}})$ retains its role as the explicit forward-reconstruction refinement.

Height. The height constituent is modeled on the propinquity for metric spectral triples [15], which measures the Lipschitz-distortion of the tunnel maps with respect to the Lipschitz seminorms, *not* the operator norm of the maps themselves. Specifically,

$$\mathrm{height}_B \;:=\; \sup_{\|f\|_{\mathrm{Lip}} \leq 1} \Big| \, \|f\|_{\mathrm{Lip}} - \|B_{n_{\max}}(f)\|_{\mathrm{Lip}}^{\mathcal{O}_{n_{\max}}} \Big|. \tag{27}$$

[REFUTED AS STATED — withdrawn 2026-09-03.] The argument printed here until that date concluded $\mathrm{height}_B \leq \gamma_{n_{\max}}$ and $\mathrm{height}_P = 0$. Both are false, by a one-line witness. $B_{n_{\max}}$ is a finite-band reconstruction — its multiplier envelope is $N \leq 2n_{\max} - 1$ — so at every finite cutoff the unit-Lipschitz ball contains f with $B_{n_{\max}}(f) = 0$, and for any such f the quantity in (27) equals $\|f\|_{\mathrm{Lip}} = 1$. Hence

$$\mathrm{height}_B \;\equiv\; 1 \quad \text{at every } n_{\max}, \tag{28}$$

and $\mathrm{height}_B \leq \gamma_{n_{\max}}$ fails whenever $\gamma_{n_{\max}} < 1$, i.e. for every $n_{\max} \geq 6$ ($\gamma_6 = 0.98958$, $\gamma_7 = 0.88277$). The same witness gives $\mathrm{height}_P \neq 0$. Two steps of the withdrawn chain are independently invalid: the inequality $\|f\|_{\mathrm{Lip}} - \|[D_{\mathrm{CH}}, M_{K_{n_{\max}} * f}]\|_{\mathrm{op}} \leq \|f\|_{\mathrm{Lip}} - \|K_{n_{\max}} * f\|_{\mathrm{Lip}}$ runs the wrong way, M being the *compressed* multiplier of §2.3 and compression contracting; and the good-kernel estimate bounds $\|K_{n_{\max}} * f - f\|_\infty$, not a difference of Lipschitz seminorms, which would require a second-order modulus.

Scope of the withdrawal. What falls is this lemma and the L5 route to Theorem 4.1. Theorem 1.7 is untouched: it uses only reach-type estimates and no height constituent, so the convergence statement — and with it the WH1 keystone — stands on that route. A height-controlled assembly for this pair is an open problem.

The assembly this proof used to close with is withdrawn with the height constituents. It read

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\leq\; \max\big(\gamma_{n_{\max}}, \gamma_{n_{\max}}, \gamma_{n_{\max}}, 0\big) \;=\; C_3 \cdot \gamma_{n_{\max}},$$

whose third and fourth arguments are the refuted height$_B \le \gamma_{n_{\max}}$ and height$_P = 0$. **The conclusion** (26) **is therefore not established by this route.** What survives is Theorem 1.7, which bounds the same state-space GH distance by $\gamma_{n_{\max}}$ using reach-type estimates alone and no height constituent — so the *statement* of Theorem 4.1 stands, by that route, and the WH1 keystone with it. □

Remark 3.13 (On the height term)**.** The height in (27) is the spectral-triple-level analog of the height in Latrémolière [15, §4], which measures the Lipschitz-seminorm-comparison error introduced by the tunnel maps. The naive bound $\|B_{n_{\max}}(f)\|_{\mathrm{op}} \le \|f\|_{L^\infty} \le 2\pi$ (from L4(b) plus the S^3 diameter in the dual-Coxeter metric) is not the relevant height for the spectral-triple propinquity: the operator norm of $B_{n_{\max}}(f)$ enters the height for the quantum compact metric space propinquity [16] but not for the spectral triple version taken as the model here. Until 2026-09-03 the reduction of the height to $\gamma_{n_{\max}}$ [retracted 2026-09-04: l5-height-bound-achieved] via the Lipschitz-comparison form was presented as what gives the rate-controlled bound of Theorem 4.1; that reduction is refuted ((28)), and the bound comes from Theorem 1.7.

4 Main theorem

Theorem 4.1 (Main theorem; rate-controlled restatement)**.** ***[INTERNAL THEOREM]*** *Unconditionally (Theorem 1.7), the truncated Camporesi–Higuchi spectral triples $\mathcal{T}_{n_{\max}}$ converge to the round-S^3 Camporesi–Higuchi spectral triple $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance [8], under the translation-seminorm metrization of § 1.1:*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; \gamma_{n_{\max}}, \qquad \gamma_{n_{\max}} = \frac{4\log n_{\max}}{\pi\, n_{\max}} + O\big(1/n_{\max}\big). \tag{29}$$

[OBSERVATION] *The asymptotic constant is the sphere-volume quotient* $2\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ *at unit radius (numerically* $\mathrm{Vol}(S^2)/\pi^2$*); it is* not *the Hopf fibration's base-to-total ratio, whose base carries half the radius (§3.2, volume-ratio note).* ***[PANEL-VERIFIED]*** *A uniform bound* $\gamma_{n_{\max}} \le 6\log n_{\max}/n_{\max}$ *holds for all* $n_{\max} \ge 2$*.*

Proof. Lemma 3.1 supplies the operator-system substrate; Lemma 3.4 the central spectral Fejér kernel with closed-form $\gamma_{n_{\max}}$ and asymptotic rate; Lemma 3.7 the Lipschitz comparison constant $C_3 = 1$; Lemma 3.10 the Berezin reconstruction with the four properties a Latrémolière tunneling pair would require; and—in the *withdrawn* route only—Lemma 3.12 the assembly into the state-space GH bound.

Which route establishes this. Not the L5 assembly: its two height constituents are refuted ((28)), so (26) is *not* established by that route — as the withdrawal in Lemma 3.12 states, and as an earlier version of this proof failed to say, hedging instead only that the route "inherits the panel scope of Lemma 3.7". The bound comes from Theorem 1.7, which bounds the same state-space GH distance by $\gamma_{n_{\max}}$ using reach-type estimates alone and no height constituent, and which uses neither Lemma 3.7 nor Lemma 3.12. Combining that bound with the asymptotic and uniform rates of Lemma 3.4(d.ii)–(d.iii) gives (29). □

A companion proposition identifies the state-space GH limit explicitly.

Proposition 4.2 (Limit identification)**.** ***[INTERNAL THEOREM]*** *The state-space GH limit of the sequence $(\mathcal{T}_{n_{\max}})_{n_{\max}\ge 1}$ is*

$$\lim_{n_{\max}\to\infty} S(\mathcal{O}_{n_{\max}}) \;=\; \big(P(S^3),\, d_{\mathrm{Wass}}\big),$$

where $P(S^3)$ is the space of Borel probability measures on S^3 and d_{Wass} is the Wasserstein–Kantorovich distance with respect to the round-S^3 geodesic distance.

Proof sketch. Theorem 4.1 guarantees $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \to 0$ in the state-space GH distance, so the limit is the continuum state space. By Kantorovich–Rubinstein duality, the Connes distance on $\mathcal{T}_{S^3}$ coincides with d_{Wass} on $P(S^3)$. The L4(c) approximate-identity property pins this to the actual limit (rather than an isometry class). This is the same limit identification as in van Suijlekom [8] and Gaudillot-Estrada–van Suijlekom [7] for the scalar case. □

5 Discussion

5.1 Why SU(2) was harder than the torus

The flat-torus result of Leimbach–van Suijlekom [17] proceeds by Schur multiplier analysis on the abelian dual $\mathbb{Z}^d$, where the Fejér kernel is a tensor product of one-dimensional Fejér kernels and the cb-norm is computed via the standard abelian-group identity $\|T\|_{\mathrm{cb}} = \|\widehat{T}\|_\infty$. Three substantive complications arise on SU(2):

(i) Non-abelian Peter–Weyl. On SU(2) the Plancherel decomposition is not over $\mathbb{Z}^d$ but over half-integer weights $j \in \frac{1}{2}\mathbb{N}_{\geq 0}$ with isotypic blocks of dimension $2j+1$. The natural Plancherel weight $\sqrt{2j+1}$ is required to match the Fock-shell Peter–Weyl bijection $n = 2j+1$, and this is what produces the natural-coefficient kernel of Definition 3.3. (For central multipliers the cb-norm computation is elementary — block-scalar action — on any compact group; the genuine non-abelian difficulty is the Clebsch–Gordan structure of the multiplier symbol, not the cb-norm.)

(ii) No Kähler structure on S^3. The standard Berezin reconstruction of Hawkins [5] uses the holomorphic Toeplitz form on a Hardy space, which requires a compatible almost-complex structure. $S^3 = \mathrm{SU}(2)$ has no such structure (this is the standard fact that S^3 has trivial tangent bundle but no integrable almost-complex structure). The natural alternative is the central-spectral-Fejér-kernel form (23), which is the abelian-compact-group analog of the Connes–van Suijlekom [4] Fejér reconstruction on S^1, lifted to the non-abelian setting via the Peter–Weyl decomposition.

(iii) The SO(4) selection rule. [OBSERVATION] The Lipschitz comparison (20) is as tight as it is because of the multiplier sparsity enforced by the SO(4) selection rule on the Avery 3-Y integral; we do not prove the bound is attained. The torus analog has no comparable sparsity: on $\mathbb{T}^d$, multiplier matrices are full Toeplitz matrices, and the Lipschitz comparison ratio is bounded by the top mode index $|k|$ rather than by the cleaner $\sqrt{(N-1)/(N+1)}$ form.

The cumulative effect is that the SU(2) rate is one log factor slower than the torus rate ($\log n/n$ vs $1/n$). This is the standard slowdown of non-abelian compact groups relative to flat tori in Cesaro/Fejér theory. **[OBSERVATION]** In the dual-Coxeter normalisation the asymptotic constant $4/\pi$ is twice the circle Fejér first-moment constant $2/\pi$; on the unit S^3 the two coincide, the factor 2 being the metric scale (Remark 3.5, 2026-09-03); we read the SU(2) value as the sphere-volume quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ at unit radius, i.e. $2\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ in the dual-Coxeter normalisation — both spheres of radius 1, so this is not the Hopf base-to-total ratio. (An earlier version of this sentence stated the constant as "identical on both sides"; withdrawn 2026-09-02.)

5.2 Marcolli–van Suijlekom gauge-network framework

The convergence theorem 4.1 provides the metric foundation for almost-commutative spectral triple constructions hosted on S^3. The Marcolli–van Suijlekom gauge-network framework [18], with the Perez-Sanchez [27] continuum-limit correction, builds finite spectral triples from algebra-valued data on graphs; **[INTERNAL THEOREM]** the present convergence result identifies one such construction — the Connes–van Suijlekom truncation of $\mathcal{T}_{S^3}$ — as metric-asymptotically the round-S^3 Camporesi–Higuchi geometry. This anchors the gauge-

network framework to a continuum geometry whose Dirac spectrum is independently known and whose spectral-action expansion is computable.

The natural next question — whether the almost-commutative extension $\mathcal{T}_{S^3} \otimes \mathcal{T}_F$ for a finite spectral triple $\mathcal{T}_F$ admits an analogous convergence statement — is open. **[MEASURED]** Inner-fluctuation analysis of the truncated AC extension at finite $n_{\max}$ has been carried out in supporting work by the present author and reproduces the expected gauge content under the Marcolli–van Suijlekom mechanism (`tests/test_almost_commutative.py`, Sprint H1; Paper 32 § Sprint H1), but the present paper restricts attention to the geometric (single-factor) state-space GH statement.

5.3 Obstruction to a Riemannian–Hardy tensor product

A natural extension of Theorem 4.1 would aim at the tensor-product spectral triple $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\text{Bargmann}}$ obtained by tensoring the present Camporesi–Higuchi triple on S^3 with a Bargmann-segal-type spectral triple on S^5 in the holomorphic Hardy sector. **[OBSERVATION]** This extension is structurally obstructed at the framework level — the obstruction is argued from the categorical mismatch below, not proved as a theorem. The S^3 factor is Riemannian with KO-dimension 3 and a second-order Laplace–Beltrami structure; the S^5 factor is a Hardy-sector first-order complex Euler operator construction [22] which does not fit standard NCG Dirac axioms. The two factors live in different categories of NCG construction, and there is no published prescription for tensoring them. We have flagged this elsewhere as the four-layer Coulomb/HO asymmetry resurfacing at the spectral-triple level. A proper Riemannian–Hardy propinquity theory is not in the literature and would require new framework development beyond the scope of the present paper.

5.4 Open questions

(i) Quantitative tightness. Theorem 4.1 gives the asymptotic constant $4/\pi$ in $\gamma_{n_{\max}} \sim (4/\pi)\log n_{\max}/n_{\max}$ but not the next-order constant. **[MEASURED]** Numerical doubling-estimator data suggests $n_{\max}\gamma_{n_{\max}} = (4/\pi)\log n_{\max} + b + O(1/n_{\max})$ with $b \approx 4.10$, whose closed form is undetermined at the level of the present analysis.

Update (2026-05-16): **[MEASURED]** the value b (rotation-angle normalisation) has been refined to 22 decimal digits (no frozen test in this repository computes b beyond six digits; the figure is a Paper 40 cross-reference) as $b = 4.1093214674898756204229\ldots$ (the original 80-digit string circulated in working memos prior to this paper was contaminated from digit 12 onward by float-precision panel points; corrected via re-extraction at ≥ 400 dps in a follow-on sprint, see [23] for details). PSLQ identification has been attempted at coefficient ceiling 10^8 against an extended L-derivative basis $(\zeta'(s), \beta'(s), \eta'(s), L'(2, \chi_q)$, Stieltjes constants, Glaisher–Kinkelin $\log A$) with NULL result. **[OPEN]** The closed form of b remains open, and the constant joins a small documented list of natural-but-unidentified constants in this programme, alongside the numerical coincidence formula studied in [24].

(ii) Lorentzian extension. The Camporesi–Higuchi spectral triple is Riemannian, with KO-dimension 3 (mod 8). On the Riemannian side, the modular-Hamiltonian structure of the truncations is closed at the operator-system level: **[MEASURED]** the modular Hamiltonian on $\mathcal{T}_{n_{\max}}$ satisfies the period closure $\sigma_{2\pi}(O) = O$ bit-exactly at $n_{\max} \in \{2, 3, 4, 5\}$ for the hemispheric-wedge-restricted state and all four physical witnesses (Bisognano–Wichmann, Hartle–Hawking, Sewell, Unruh), in both the geometric realization ($K = J_{\text{polar}}$, integer spectrum, so $e^{i\cdot 2\pi n} = 1$ exactly) and the Tomita–Takesaki realization ($K_{\text{TT}} = -\log\Delta$ on the GNS Hilbert–Schmidt space), which are flow-conjugate, $\sigma_t^{\text{TT}} = \sigma_{-t}^{\alpha}$; see the four-witness companion paper [20] and [19] § VIII.F. **[OPEN]** The genuinely Lorentzian extension at signature $(3, 1)$ via Krein-space spectral triples is *open*, and is now known to be obstructed along its most natural route: compressing a Krein-self-adjoint product Dirac to the Krein-positive subspace annihilates

the spatial Dirac, so the compressed Lipschitz seminorm vanishes identically on the natural operator system and no quantum-metric structure survives. The companion Lorentzian paper [21] proves this degeneracy theorem and states the repaired target (a Toeplitz temporal multiplier algebra in the Connes–van Suijlekom circle pattern, and a Krein-compatible Lipschitz seminorm that does not compress the Dirac).

(iii) Higher-rank compact non-abelian cases. The present proof transcribes mechanically to SU(n) for $n \geq 3$ at the algebraic level (Peter–Weyl, central characters, Plancherel weights) but the Lipschitz comparison estimate (21) requires a higher-rank version of the Avery 3-Y integral and the corresponding selection rule.

Update (2026-05-16): this open question has been resolved by Paper 40 [23], which generalises Theorem 4.1 to all compact connected semisimple Lie groups with bi-invariant metric (a circle factor carries the constant $2/\pi$ of Remark 3.5, so tori are excluded). The higher-rank analog of the Avery 3-Y selection rule is the *Dirac-triangle inequality* $|D(\pi) - D(\pi')| \leq \sqrt{C(\sigma)}$ for every $\sigma \subset \pi \otimes \pi'^{*}$, which generalises Paper 38's $|n - n'| \leq N - 1$ shell-difference rule (the literal "Casimir triangle inequality" $|C(\pi) - C(\pi')| \leq C(\sigma)$ is false in general; the Dirac eigenvalue, not the Casimir, is the right ingredient). **[PANEL-VERIFIED]** The rate is *not* rank-dependent as anticipated above: the rate constant $4/\pi$ (dual-Coxeter normalisation) is reported *universal* across the class, by a Plancherel-weight × Vandermonde-Jacobian cancellation theorem (Paper 40 [23] §3.2) — rigorous at rank 1 (SU(2), the present paper), with the rank-≥ 2 leg corroborated on reduced panels and a fully rank-uniform analytical proof recorded as a named gap in Paper 40. **[INTERNAL THEOREM]** Asymptotic-tight $C_3(G) = 1$ at all ranks follows from the PRV-summand bound (Kumar 1988, Invent. Math. 93) together with a Brauer–Klimyk signed-sum closure for the interior summands, as proved in Paper 40 §3.3 (Lemma L3 there; the attribution to a 1990 Vinberg paper carried no locator and is withdrawn, 2026-09-03).

A Detailed derivation of Lemma 3.4(d.ii)

[INTERNAL THEOREM + MEASURED] This appendix supplies the sum-rule / Euler–Maclaurin derivation (the "Stein–Weiss sharpening" in the corpus's naming, after the real-line kernel estimates of [28]) of the asymptotic $\lim_{n\to\infty} n\gamma_n/\log n = 4/\pi$ that appears as a sketch in the proof of Lemma 3.4. Step 3 isolates the two sub-coefficients (2 and -1) whose net is the logarithm's coefficient; each sub-coefficient and the net are pinned by independent doubling estimators (test-backed), which is what closes the constant. The argument is a standard asymptotic analysis of the closed-form sum-rule (16); we record it here in full because the referee may want to see the leading-order constant pinned without having to reconstruct it.

Setup. Recall the closed form of Lemma 3.4(d.i):

$$\gamma_n \;=\; \pi \;-\; \frac{4T_n}{\pi Z_n}, \qquad Z_n = \frac{n(n+1)}{2}, \qquad T_n = \sum_{\substack{1\leq k_1,k_2\leq n\\ k_1+k_2 \text{ odd}}} \sqrt{k_1 k_2}\Big[\frac{1}{(k_1-k_2)^2} - \frac{1}{(k_1+k_2)^2}\Big]. \tag{30}$$

The asymptotic claim of Lemma 3.4(d.ii) is equivalent to

$$\frac{T_n}{Z_n} \;=\; \frac{\pi^2}{4} \;-\; \frac{\log n}{n} \;+\; O\Big(\frac{1}{n}\Big), \tag{31}$$

since $\gamma_n = \pi - (4/\pi)(T_n/Z_n) = (4/\pi)(\log n/n) + O(1/n)$.

Step 1: Diagonal-dominant decomposition. Write $k_2 = k_1 + d$ with d odd, $d \in \{\pm 1, \pm 3, \ldots\}$, $k_1 \in \{1, \ldots, n\}$, $k_1 + d \in \{1, \ldots, n\}$. By the $k_1 \leftrightarrow k_2$ symmetry of T_n, we may take $d > 0$ and

double:

$$T_n \;=\; 2\sum_{\substack{d>0\\ d\,\mathrm{odd}}}\sum_{a=1}^{n-d}\sqrt{a(a+d)}\Big[\frac{1}{d^2}-\frac{1}{(2a+d)^2}\Big].$$

Step 2: Leading-order $\pi^2/4$. The term $1/d^2$ dominates. Using $\sqrt{a(a+d)} = a\sqrt{1+d/a} = a + d/2 - d^2/(8a) + O(d^3/a^2)$ and $\sum_{a=1}^{n-d} 1 = n-d$,

$$\sum_{a=1}^{n-d}\sqrt{a(a+d)}\cdot\frac{1}{d^2} \;=\; \frac{1}{d^2}\Big[\frac{(n-d)(n-d+1)}{2}+\frac{d(n-d)}{2}+O(d\log n)\Big] \;=\; \frac{n^2}{2d^2}+O\Big(\frac{n}{d}\Big).$$

Summing over odd d from 1 to n and using $\sum_{d\,\mathrm{odd}} 1/d^2 = \pi^2/8$ (the Catalan-Euler-style identity $\sum_{d\geq 1\,\mathrm{odd}} 1/d^2 = (1-2^{-2})\zeta(2) = (3/4)(\pi^2/6) = \pi^2/8$),

$$T_n \;=\; 2\cdot\frac{n^2}{2}\cdot\frac{\pi^2}{8}+\text{lower order} \;=\; \frac{\pi^2 n^2}{8}+\text{lower order}.$$

Dividing by $Z_n = n(n+1)/2 \sim n^2/2$ gives the leading $T_n/Z_n \to \pi^2/4$, confirming $\gamma_n \to 0$.

Step 3: Sub-leading $\log n$ (two sub-coefficients, $2-1=1$). Write $\mathrm{defect}_n := \pi^2/4 - T_n/Z_n$, so that (31) is the statement $\mathrm{defect}_n = \log n/n + O(1/n)$, i.e. that the $\log n/n$ coefficient of the defect is exactly 1 (and $\gamma_n = (4/\pi)\,\mathrm{defect}_n$ identically by (30)). Reorganise the Step-1 double sum by a, with d odd and $d \leq n-a$, and expand $\sqrt{a(a+d)} = a + d/2 + r(a,d)$. Two pieces carry a logarithm:

(A) *Triangle truncation of the d-sum.* The $a\cdot(1/d^2)$ piece is $2\sum_{a=1}^{n-1} a\,S_2(n-a)$ with $S_2(D) = \sum_{d\,\mathrm{odd}\leq D} d^{-2} = \pi^2/8 - 1/(2D) + O(D^{-2})$ (Euler–Maclaurin on the odd-$\zeta(2)$ tail). The tail contributes $-\sum_{a=1}^{n-1} a/(n-a) = -nH_{n-1} + O(n)$, so this piece alone gives defect log-coefficient 2.

(B) *The $d/2$ correction.* The $(d/2)\cdot(1/d^2)$ piece is $\sum_{a=1}^{n-1} S_1(n-a)$ with $S_1(D) = \sum_{d\,\mathrm{odd}\leq D} d^{-1} = \frac{1}{2}\log D + O(1)$, which is $\frac{1}{2}n\log n + O(n)$; it *adds* to T_n/Z_n and so reduces the defect with log-coefficient 1.

The remaining pieces — the $-1/(2a+d)^2$ bracket and the residual $r(a,d)$ — carry no $\log n$ at leading order: the net coefficient on the exact T_n is $2-1=1$. **[MEASURED]** All three statements are pinned independently by doubling (Richardson) estimators on the genuinely computed partial sums, none of which assumes the answer: the piece-(A) estimator converges to 2, the piece-(B) estimator to 1, and the estimator on the exact defect_n to 1 (`tests/test_trunk_qa_fejer_4_over_pi.py`: `test_subcoeff_A_triangle_truncation_is_two`, `test_subcoeff_B_sqrt_correction_is_one`, `test_net_defect_coefficient_converges_to_one`). Hence

$$\frac{T_n}{Z_n} \;=\; \frac{\pi^2}{4} \;-\; \frac{\log n}{n} \;+\; O\Big(\frac{1}{n}\Big),$$

which is (31). (An earlier version of this step attributed the logarithm to the $-1/(2a+d)^2$ bracket and to a boundary term in $\sqrt{a(a+d)}$, each "$-\pi^2 n\log n/16$"; that bookkeeping did not compose to the displayed result and is replaced by the test-backed decomposition above, 2026-09-02.)

Step 4: Conversion to γ_n. Substituting into (30),

$$\gamma_n \;=\; \pi \;-\; \frac{4}{\pi}\Big(\frac{\pi^2}{4}-\frac{\log n}{n}+O(1/n)\Big) \;=\; \frac{4\log n}{\pi n} \;+\; O\Big(\frac{1}{n}\Big),$$

which is the asymptotic of Lemma 3.4(d.ii).

Numerical verification. **[MEASURED]** The closed form $n\gamma_n = (4/\pi)\log n + b + O(1/n)$ with $b \approx 4.109$ has been verified to high precision against the sum-rule (30) at $n \in \{2, \ldots, 1000\}$, with the doubling estimator $a_n := (2n\gamma_{2n} - n\gamma_n)/\log 2 \to 4/\pi$ reaching 1.2785 at $n = 800$ and 1.2761 at $n = 1600$ (three digits of $4/\pi$; residual shrinking $\sim$1.85$\times$ per doubling) (`tests/test_trunk_qa_fejer_test_gamma_doubling_converges_to_4_over_pi`, `test_su2_doubling_estimator_at_n800` and the slow-marked `test_su2_doubling_estimator_at_n1600`; the sum-rule itself is checked against Gaussian quadrature of the kernel moment in `tests/test_central_fejer_su2.py`).

Relation to the circle Fejér estimate. The constant $4/\pi$ is *twice* the constant of the classical estimate $\frac{1}{2\pi}\int_{\mathbb{T}^1} F_n(\theta)\,|\theta|\,d\theta \sim (2/\pi)\log n/n$ for the probability-normalised Fejér kernel on the circle (Zygmund [29] Vol. I, Ch. III; Remark 3.5, where the circle constant is computed in closed form and measured). The $\sin^2(\chi/2)$ factor in the SU(2) conjugacy-class measure and the $\sqrt{2j+1}$ Plancherel weighting of Definition 3.3 together change the kernel and the constant while preserving the leading $\log n/n$ behaviour. **[OBSERVATION]** On the circle the constant is $2\mathrm{Vol}(S^0)/\mathrm{Vol}(S^1) = 2/\pi$; on SU(2) it is $\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = 2/\pi$ on the unit S^3 and $2\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = \mathrm{Vol}(S^2)/\pi^2 = 4/\pi$ in the dual-Coxeter normalisation; no common "2Vol(base)/Vol(group)" form is claimed (2026-09-03), and no derivation of one constant from the other is given. (An earlier version of this paragraph stated the circle constant as $4/\pi$ and the SU(2) constant as "preserved across this transition"; withdrawn 2026-09-02.)

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